

SHIVAM KUMAR SINGH

PYTHON & GEN AI DEVELOPER



PROFILE

Python & GenAI Developer with over 5.5 years of experience in developing scalable web applications and integrating Generative AI solutions. Proficient in Python, FastAPI, Django, Flask, with expertise in LLMs (OpenAI GPT, LLaMA, LangChain, LangGraph, RAG) and practical deployment on AWS (EC2, S3, RDS, CloudWatch) for secure, high-performance systems.



WORK EXPERIENCE

PwC 08/2022 - Present
Associate | Python and Gen AI Developer

- Integrated Azure OpenAI GPT-4 via LangChain to automate conversation auditing for compliance, quality assurance, and actionable insights.
- Implemented RAG pipelines by integrating external knowledge sources, enhancing chatbot responses.
- Built a FastAPI backend with SQLAlchemy, Redis, and Pydantic, deployed using Nginx, delivering 35% faster response times, and improved data accuracy.
- Implemented Azure Speech-to-Text (STT) with Azure SDK for Python and diarization support, enabling real-time transcription with speaker identification.
- Implemented an agentic AI system using LangGraph and BrightData SDK, automating data extraction from external sources.
- Applied prompt engineering techniques to optimise GPT-4 and LLaMA 2 responses, improving accuracy, consistency, and compliance in outputs.

Amtz Limited 08/2021 - 08/2022
Software Developer

- Developed scalable web applications using Django to automate financial report generation, integrating data from 5+ disparate sources and reducing manual effort by 40%, while improving report accuracy and delivery speed.
- Designed and implemented RESTful APIs using Django REST Framework to enable seamless integration with internal systems and automate reporting workflows.
- Managed PostgreSQL databases, writing optimized queries to improve performance, reliability, and data consistency.
- Deployed and maintained applications on AWS services including EC2, S3, and RDS, and implemented monitoring and alerting using Amazon CloudWatch.
- Collaborated with financial analysts and cross-functional stakeholders to gather requirements, validate reporting logic, and deliver accurate, production-ready solutions.

CONTACT

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SKILLS

- Programming Languages : Python
- Web Development : FastAPI, Django, Flask
- Gen AI : Open AI, Llama, LangChain, LangGraph, ChromaDB, Retrieval-Augmented Generation (RAG)
- NLP : NLTK, SpaCy, TextBlob
- Data Preprocessing : Pandas, Pyspark
- Data Visualization : Matplotlib, Seaborn, Plotly
- AWS (EC2, S3, RDS CloudWatch)
- Azure (ADLS, Azure Speech to Text, Azure Blob Storage)
- Database Management : PostgreSQL, Mysql, MongoDB, Redis
- Workflow Orchestration : Apache Airflow, Cron, Celery
- Deployment: Nginx, Docker
- Version Control : GitHub, SVN
- Frontend : HTML, CSS, JavaScript

LANGUAGES

- English (Fluent)
- Hindi (Fluent)



Chadha Industries Private Limited07/2020 - 08/2021

Software Developer

- Built a Django backend for an internal financial data analysis tool, enabling secure storage and retrieval of large-scale financial datasets.
- Developed REST APIs with Django REST Framework to support data access, reporting, and integration with internal systems.
- Implemented data preprocessing workflows using Pandas to clean and transform raw financial records into analysis-ready formats.
- Deployed the application on AWS EC2 with Nginx and Unicorn, achieving high availability and optimised performance.



Jaarwis Technology Private Limited12/2019 - 06/2020

Software Developer

- Developed and modified RESTful APIs using Flask to support core application features, ensuring secure and efficient data exchange across internal modules.
- Implemented and scheduled cron jobs to automate recurring backend tasks such as data synchronisation, cleanup, and email notifications via SMTP.
- Optimised backend performance by debugging, testing, and refactoring Flask components, improving application reliability and maintainability.



EDUCATION



BTech2015 - 2019

Maharaja Agrasen Institute Of Technology, GGSIPU, Delhi

CGPA : 7



Class 12th in PCM2014 - 2015

St. Crispin Senior Secondary School, CBSE, Gurgaon

Grade - 84.4 %



Class 10th2012 - 2013

S D Memorial High School, CBSE, Gurgaon

CGPA : 9.4